

Business Requirement Document

AI-Powered Customer Support Ticket Triage
SaaS Delivery Simulation for Customer Experience Operations

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Client context	UrbanCart - Mock E-commerce Enterprise
Document type	Portfolio Business Requirement Document
Confidentiality	Academic / Portfolio Simulation

No real customer data, production deployment, or live client implementation is represented in this document.

Document Overview

This Business Requirement Document defines the business need, stakeholder expectations, scope, requirements, risk controls, and measurable outcomes for an AI-enabled customer support ticket triage workflow. The document is written as a professional SaaS delivery simulation for a mock e-commerce enterprise and is intended to demonstrate requirement analysis, implementation planning, and delivery documentation capability.

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1. Executive Summary

UrbanCart, a mock e-commerce enterprise, receives a high volume of customer support tickets across common issue types such as refunds, delivery delays, payment failures, cancellations, damaged products, login issues, and general service queries. The current manual triage approach requires support agents to read each message, interpret the issue, decide urgency, and route the request to the correct team. This creates operational friction, inconsistent categorization, missed escalations, and slower first response times.

The proposed solution is an AI-enabled SaaS workflow that assists with ticket classification, priority assignment, and routing. The solution does not claim to train or deploy a real artificial intelligence model; instead, it documents a realistic delivery approach using explainable classification logic, structured business requirements, and implementation-ready artifacts.

The purpose of this BRD is to define the business case and delivery expectations so that product, technology, support, implementation, and customer success teams can align on the target workflow before product configuration or development begins.

Business Need	Proposed Capability	Expected Outcome
Reduce manual ticket sorting and inconsistent issue tagging.	AI-assisted classification into standardized ticket categories.	Improved response consistency, faster routing, and better support visibility.
Improve urgent case handling and escalation visibility.	Priority assignment using issue type, urgency indicators, and escalation language.	High-priority tickets can be identified and acted on faster.
Create a clear implementation baseline for internal teams.	BRD, PRD, user stories, demo script, training guide, and sample data.	Better cross-functional alignment between business and technical teams.

2. Business Context and Current-State Assessment

UrbanCart's support operations depend on timely and accurate handling of customer issues. As ticket volume grows, manual triage becomes a bottleneck. The support team must repeatedly interpret similar customer messages, select the correct category, identify urgency, and send the case to the right operational team. Without a standardized triage workflow, similar issues can receive different treatment depending on agent experience, queue pressure, or unclear routing rules.

The current-state process is workable at low volume but does not scale well for enterprise-grade customer experience operations. A SaaS-based triage layer can improve repeatability and create a consistent foundation for future automation, analytics, and service-level monitoring.

Area	Current-State Observation	Business Impact
Ticket Intake	Customer messages arrive as unstructured text across issue types.	Agents spend time interpreting and sorting requests manually.
Categorization	Issue labels are applied manually and may vary by agent.	Reporting accuracy and routing consistency are reduced.
Prioritization	Urgency is identified through manual judgment.	Payment and escalation cases may not be addressed quickly enough.
Routing	Tickets are routed based on category understanding and agent knowledge.	Misrouted tickets create rework and longer resolution cycles.
Visibility	Managers may lack a consolidated view of urgent or repeated issues.	Operational decisions become reactive rather than proactive.

3. Problem Statement and Opportunity

UrbanCart requires a standardized and scalable mechanism to classify, prioritize, and route incoming customer support tickets. The absence of a structured triage workflow leads to inconsistent handling, slower response, inefficient team allocation, and limited managerial visibility into customer pain points.

The opportunity is to simulate an AI-enabled SaaS delivery workflow that can convert unstructured customer messages into operationally useful ticket attributes: category, priority, routing team, and manual review flag. This creates a clear business-to-technology bridge for solution delivery.

Primary business pain points

- High manual effort in reading, interpreting, and sorting support messages.
- Inconsistent categorization of similar issues across support agents.
- Delayed detection of urgent payment, damaged product, or repeated complaint cases.
- Misrouting between billing, logistics, technical support, returns, and general support teams.
- Limited structured data for support reporting, quality monitoring, and continuous improvement.

4. Business Objectives and Success Outcomes

The business objectives focus on operational efficiency, customer experience consistency, and delivery readiness. The workflow should improve the quality of ticket intake and provide a repeatable foundation for product, technology, support, and implementation teams.

Objective ID	Business Objective	Success Outcome
OBJ-01	Classify tickets into standardized categories.	Support teams can identify the issue type without manual interpretation for every ticket.
OBJ-02	Assign priority based on issue type and urgency.	Critical issues are surfaced faster for action.
OBJ-03	Route tickets to the correct operational team.	Reduced rework caused by incorrect handoffs.
OBJ-04	Support manual review for ambiguous tickets.	The workflow remains reliable even when classification confidence is low.
OBJ-05	Create delivery documentation for implementation readiness.	Business, product, and technology teams have a shared reference for execution.
OBJ-06	Enable future reporting and continuous improvement.	Ticket trends and recurring customer issues become easier to analyze.

5. Stakeholder and User Group Analysis

Successful SaaS delivery depends on alignment between user groups, business owners, product teams, and implementation stakeholders. The following matrix defines stakeholder responsibilities and expected involvement.

Stakeholder	Primary Responsibility	BRD Relevance
Support Agents	Review and resolve categorized tickets.	Need clear category, priority, and routing visibility.
Support Manager	Monitor ticket queues, escalations, and operational performance.	Needs reporting-ready data and priority visibility.
Implementation Team	Translate business requirements into configurable workflows and delivery artifacts.	Owens requirement clarity, open questions, handover readiness, and execution coordination.
Product Team	Review feature scope, workflow logic, and product feasibility.	Uses BRD inputs to prepare or validate PRD-level requirements.

Stakeholder	Primary Responsibility	BRD Relevance
Technology Team	Validate classification logic, data handling, integration assumptions, and workflow behavior.	Needs functional clarity, data expectations, and edge-case handling.
Customer Success Team	Support demos, training, adoption, and client communication.	Needs demo script, training guide, and clear business value narrative.

6. Scope Definition

The scope defines what will and will not be included in the delivery simulation. This prevents overclaiming and keeps the portfolio project aligned with realistic entry-level AI delivery responsibilities.

In Scope	Out of Scope
Ticket category classification for predefined support issue types.	Training or deploying a production-grade AI model.
Priority assignment using documented decision rules.	Processing real customer data or personally identifiable information.
Routing logic between issue categories and support teams.	Live SaaS platform deployment or enterprise integration.
Manual review path for unclear or uncategorized tickets.	Payment gateway, CRM, logistics, or identity-provider integration.
BRD, PRD, user stories, implementation plan, demo script, training guide, and sample data.	Production-level security controls, compliance certification, or service-level contract enforcement.

Scope control principle: The project should be presented as an implementation and documentation simulation. It should not be represented as a live client engagement, a deployed SaaS module, or a trained machine learning product.

7. Future-State Process Overview

The future-state workflow introduces a structured triage layer between ticket intake and agent resolution. The workflow converts a customer message into business-operational attributes that can be used by support agents and managers.

Step	Workflow Activity	Output
1	Customer submits a support ticket through the support channel.	Unstructured text message is received.
2	SaaS platform captures ticket details and sends message to triage workflow.	Ticket is ready for classification.
3	AI-assisted workflow identifies likely issue category.	Category such as Payment Issue, Refund, or Delivery Delay is assigned.
4	Priority logic evaluates urgency and escalation signals.	Priority is set as High, Medium, or Low.
5	Routing logic maps ticket category to an operational team.	Ticket is assigned to the correct support queue.
6	Support agent reviews category, priority, and message context.	Agent can respond or mark ticket for manual review.
7	Manager monitors queue health and urgent case volume.	Operational visibility improves.

Workflow summary

Customer Ticket -> SaaS Intake -> AI-Assisted Classification -> Priority Assignment -> Team Routing -> Agent Review -> Customer Response -> Manager Visibility

8. Business Requirements

The following requirements define the business capabilities expected from the solution. Each requirement is written in a way that can later be converted into PRD-level functional requirements, user stories, acceptance criteria, and implementation tasks.

ID	Business Requirement	Priority	Business Rationale	Acceptance Signal
BR-01	The system should classify incoming tickets into predefined issue categories.	High	Standardized classification reduces agent interpretation effort and improves reporting quality.	Each ticket receives a visible category from the approved category list or is marked for manual review.
BR-02	The system should assign a priority level based on issue type, urgency, and escalation signals.	High	Urgent issues must be surfaced early to protect customer experience and reduce escalation risk.	Every ticket displays High, Medium, or Low priority with clear rule-based reasoning.
BR-03	The system should route tickets to the correct support team based on ticket category.	High	Correct routing reduces rework and shortens resolution cycles.	Each category maps to a defined support team or fallback queue.
BR-04	High-priority tickets should be flagged for faster support handling.	High	Payment issues, damaged products, repeated complaints, and escalation language require quicker attention.	High-priority tickets are visibly flagged in the support queue.
BR-05	Support agents should be able to view ticket category, priority, and routing team.	Medium	Agents need context before responding to customers.	Ticket view includes category, priority, routing team, and customer message.
BR-06	The workflow should support manual review for uncategorized or ambiguous tickets.	Medium	Manual fallback protects service quality when automated logic is uncertain.	Unclear tickets are routed to manual review or general support.
BR-07	Support managers should be able to monitor high-priority and manually reviewed tickets.	Medium	Operational leaders need visibility into urgent and ambiguous cases.	Manager view can identify priority and manual review volume.
BR-08	The implementation documentation should remain clear enough for product, technology, and support teams.	High	Delivery quality depends on shared understanding across teams.	BRD, PRD, user stories, demo script, and training guide are aligned and complete.

9. Data, Rules, and Decision Logic

The simulation uses sample support tickets and explainable classification rules. The classification logic is intentionally transparent so that business users can understand why a ticket is assigned to a category or priority level.

Category	Typical Customer Signal	Default Routing Team	Default Priority
Refund	Refund request, refund status, amount not received.	Billing & Refund Team	Medium
Delivery Delay	Order late, not received, missed delivery estimate.	Logistics Team	Medium
Payment Issue	Payment failed, amount deducted, order not placed.	Payments Team	High
Damaged Product	Broken, defective, damaged, replacement needed.	Returns & Replacement Team	High
Cancellation	Cancel order, order placed by mistake.	Order Management Team	Medium
Login Issue	Cannot log in, password reset, account access.	Technical Support Team	Low

Category	Typical Customer Signal	Default Routing Team	Default Priority
General Query	Address change, order status, basic support query.	General Support Team	Low

Priority	Trigger Conditions	Operational Treatment
High	Payment failure, money deducted, damaged product, repeated complaint, urgent escalation language, strong negative sentiment.	Surface early, route immediately, monitor by manager.
Medium	Refund pending, delivery delayed, cancellation requested, moderate urgency.	Route to correct queue and handle within normal support process.
Low	General query, account help, order status request, basic information request.	Route to standard support queue.

10. Reporting, Governance, and Acceptance Criteria

For a SaaS delivery workflow to be considered ready for demonstration, it should provide clear visibility into ticket handling outcomes and support a simple governance process for changes in categories, priority rules, and routing logic.

Metric Area	Recommended Measurement	Business Purpose
Classification Coverage	Percentage of tickets assigned to a predefined category.	Shows how well the workflow handles expected issue types.
Manual Review Rate	Percentage of tickets requiring manual review.	Identifies gaps in category rules or unclear ticket language.
High-Priority Ticket Volume	Count of tickets flagged as High priority.	Helps managers track urgent customer issues.
Routing Accuracy	Percentage of tickets routed to the correct team after review.	Measures quality of category-to-team mapping.
Agent Efficiency	Estimated reduction in manual sorting effort.	Indicates operational benefit for support users.

Acceptance criteria

- Ticket categories are clearly defined and documented.
- Priority levels are defined with understandable trigger conditions.
- Routing logic maps each category to a support team.
- Manual review fallback is included for unclear tickets.
- BRD, PRD, user stories, implementation plan, demo script, training guide, and sample data are internally consistent.
- The project can be explained as a SaaS delivery simulation without claiming real deployment or live client delivery.

11. Assumptions, Dependencies, Constraints, and Risks

The delivery simulation relies on defined assumptions and clearly documented boundaries. These items are important because they show practical awareness of implementation risk and delivery governance.

Type	Statement
Assumption	Customer tickets are represented as text messages in the sample dataset.
Assumption	Ticket categories are predefined and approved before workflow configuration.
Assumption	Classification is represented through explainable rule-based logic for simulation purposes.
Dependency	The PRD and user stories should align with this BRD before demo preparation.

Type	Statement
Dependency	Sample ticket data should cover all target categories.
Constraint	No real customer data or production system is used.
Constraint	The project does not include model training, platform deployment, or enterprise integration.

Risk	Impact	Mitigation	Owner
Incorrect classification	Ticket may be routed to the wrong team.	Use manual review fallback and maintain clear category definitions.	Implementation / Product
High manual review rate	Workflow may not reduce agent effort enough.	Refine categories, review sample data, and update triage rules.	Product / Support
Unclear stakeholder expectations	Delivery artifacts may not match business priorities.	Maintain BRD, PRD, user stories, and open questions as shared references.	Implementation
Overclaiming project capability	Misrepresentation risk in portfolio context.	State clearly that this is a simulation and not a live AI deployment.	Project Owner
Routing rule gaps	Some cases may not map cleanly to a team.	Add fallback queue and periodic category review.	Support Manager

12. Implementation Readiness and Handover Criteria

The delivery simulation is ready for portfolio submission when the documentation set provides enough context for a reviewer to understand the business problem, proposed workflow, requirement logic, user stories, and demonstration approach.

Handover Item	Completion Criteria
BRD	Business problem, objectives, stakeholders, scope, requirements, metrics, assumptions, and risks are documented.
PRD	Functional requirements, non-functional requirements, user flow, routing logic, and acceptance criteria are prepared.
User Stories	User roles, expected outcomes, and acceptance criteria are written clearly.
Sample Data	Sample tickets cover the target categories and priority examples.
Demo Script	The project can be explained in a structured client-demo style.
Training Guide	Support users can understand categories, priorities, and manual review behavior.
GitHub Repository	Files are organized, readable, and labelled as an independent portfolio simulation.

Recommended interview explanation

This project can be presented as a hands-on exercise in AI delivery documentation for a SaaS customer support workflow. It demonstrates how a business problem can be converted into structured requirements, delivery artifacts, and demo-ready documentation for product, technology, support, and customer success teams.

13. Glossary

Term	Meaning in This Document
AI-Enabled Workflow	A workflow that uses automated or intelligence-assisted logic to support classification, prioritization, and routing decisions.
BRD	Business Requirement Document. Defines business problem, objectives, scope, stakeholders, and business-level requirements.

Term	Meaning in This Document
PRD	Product Requirement Document. Converts business needs into product-level functional and non-functional requirements.
SaaS	Software as a Service. A cloud-based software platform that clients use through a subscription or hosted delivery model.
Ticket Triage	The process of reviewing, categorizing, prioritizing, and routing customer support tickets.
Manual Review	A fallback path where an agent reviews a ticket that cannot be confidently categorized.
Routing Team	The internal team responsible for handling a ticket based on its issue type.

End of Business Requirement Document